# Day 1 Study Notes

Fri 13 Nov 2026 · Sprint 60 · JEE Main 2027

| Part | Time | What to do |
|---|---|---|
| **Study** | 3 h 30 min | Physics: Electromagnetic waves (75 min) · Maths: Sets and relations (75 min) · Chemistry: Biomolecules (60 min) |
| **Exam** | 1 h 15 min | The Day 1 exam in the Sprint 60 app: 30 questions, timed, no notes. |
| **Review** | 1 h 15 min | Open every wrong and skipped question in the app, read its solution, tag the reason, then use Retry my mistakes. |

**How to use each chapter note.** Read it once without stopping. Cover the formulas and write them from memory. Work the solved examples on paper before reading the working. Then read the NCERT section listed at the end.

# Physics · Electromagnetic waves

About one question in most papers. Almost always a direct formula or a fact about the spectrum, so this is one of the cheapest four marks in Physics.

## 1. Displacement current

A changing electric field produces a magnetic field, just as a current does. Maxwell added this to Ampère's law as the displacement current:

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}, \qquad \oint \vec{B} \cdot d\vec{l} = \mu_0 \left( I_c + \varepsilon_0 \frac{d\Phi_E}{dt} \right)$$

- While a capacitor charges, there is conduction current $I_c$ in the wires and displacement current $I_d$ between the plates, and $I_d = I_c$.
- For a capacitor, $I_d = C \dfrac{dV}{dt}$.
- When the capacitor is fully charged, both are zero.

## 2. Nature of electromagnetic waves

- Produced by accelerated or oscillating charges. A charge at rest or moving with constant velocity does not radiate.
- $\vec{E}$, $\vec{B}$ and the direction of travel are mutually perpendicular, so the wave is transverse. The direction of travel is along $\vec{E} \times \vec{B}$.
- $\vec{E}$ and $\vec{B}$ oscillate in phase: they reach maximum and zero together.
- Speed in vacuum: $c = \dfrac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3 \times 10^8$ m/s. Amplitudes are related by $\dfrac{E_0}{B_0} = c$.
- Speed in a medium: $v = \dfrac{1}{\sqrt{\mu \varepsilon}} = \dfrac{c}{\sqrt{\mu_r \varepsilon_r}}$, so the refractive index is $n = \sqrt{\mu_r \varepsilon_r}$.
- They need no medium, are not deflected by electric or magnetic fields, and carry both energy and momentum.
- Frequency does not change when the wave enters a medium. Speed and wavelength do.

## 3. Equation of the wave

For a wave travelling along $+x$: $E_y = E_0 \sin(kx - \omega t)$ and $B_z = B_0 \sin(kx - \omega t)$, with $k = \dfrac{2\pi}{\lambda}$, $\omega = 2\pi\nu$ and $c = \dfrac{\omega}{k} = \nu\lambda$.

> **Solved example 1.** $B = 2 \times 10^{-7} \sin(0.5 \times 10^3 x + 1.5 \times 10^{11} t)$ T. Find the speed and the electric field amplitude.
>
> Speed $= \dfrac{\omega}{k} = \dfrac{1.5 \times 10^{11}}{0.5 \times 10^3} = 3 \times 10^8$ m/s. The plus sign means it travels along $-x$. $E_0 = cB_0 = 3 \times 10^8 \times 2 \times 10^{-7} = 60$ V/m.

## 4. Energy, intensity and pressure

| Quantity | Formula |
|---|---|
| Energy density of the electric field | $u_E = \frac{1}{2}\varepsilon_0 E^2$ |

| | |
|---|---|
| Energy density of the magnetic field | $u_B = \dfrac{B^2}{2\mu_0}$ |
| Average total energy density | $u = \frac{1}{2}\varepsilon_0 E_0^2 = \dfrac{B_0^2}{2\mu_0}$ (electric and magnetic parts are equal) |
| Intensity | $I = uc = \frac{1}{2}\varepsilon_0 E_0^2 c$ |
| Intensity from a point source of power $P$ | $I = \dfrac{P}{4\pi r^2}$ |
| Momentum carried by energy $U$ | $p = \dfrac{U}{c}$ |
| Radiation pressure | $\dfrac{I}{c}$ on an absorbing surface, $\dfrac{2I}{c}$ on a reflecting surface |

## 5. The electromagnetic spectrum

In order of increasing frequency and decreasing wavelength:

| Wave | Wavelength range | Produced by | Used for |
|---|---|---|---|
| Radio | more than 0.1 m | oscillating circuits | radio and TV broadcasting, mobile phones |
| Microwave | 0.1 m to 1 mm | klystron, magnetron | radar, microwave ovens, satellite links |
| Infrared | 1 mm to 700 nm | hot bodies, vibrating molecules | remote controls, heating, night vision; causes the greenhouse effect |
| Visible | 700 nm to 400 nm | electrons changing energy levels | sight |
| Ultraviolet | 400 nm to 1 nm | the Sun, arc lamps | sterilising water, eye surgery; absorbed by the ozone layer |
| X-rays | 1 nm to $10^{-3}$ nm | fast electrons hitting a metal target | medical imaging, studying crystals |
| Gamma rays | less than $10^{-3}$ nm | radioactive nuclei | cancer treatment |

**Traps.** In "arrange in order" questions, check whether the question asks for wavelength or frequency: the orders are opposite. The plus sign in $\sin(kx + \omega t)$ means the wave travels along the negative axis. Use $E_0/B_0 = c$ only for amplitudes or for values at the same instant. For a reflecting surface the pressure doubles.

**Read in NCERT:** Physics Class 12 Part 1, chapter "Electromagnetic Waves". It is short: read the whole chapter, including the spectrum table.

# Mathematics · Sets and relations

About one question per paper. Recent papers favour counting questions on relations: how many relations are reflexive, how many pairs must be added, how many equivalence relations exist.

## 1. Sets

- A set with $n$ elements has $2^n$ subsets and $2^n - 1$ proper subsets. The set of all subsets is the power set.
- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.
- $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$.
- $n(A - B) = n(A) - n(A \cap B)$. Symmetric difference: $n(A \,\Delta\, B) = n(A) + n(B) - 2n(A \cap B)$.
- De Morgan's laws: $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$.
- "Neither" in survey questions is $n(U) - n(A \cup B)$.

> **Solved example 1.** In a group of 100 students, 60 play cricket, 50 play football and 20 play both. How many play neither?
> $n(C \cup F) = 60 + 50 - 20 = 90$, so $100 - 90 = 10$ play neither.

## 2. Cartesian product and relations

- If $n(A) = p$ and $n(B) = q$, then $n(A \times B) = pq$.
- A relation from $A$ to $B$ is any subset of $A \times B$, so there are $2^{pq}$ relations.

## 3. Types of relation on a set $A$

| Type | Condition | Quick check |
|---|---|---|
| Reflexive | $(a, a) \in R$ for every $a \in A$ | every element is paired with itself |
| Symmetric | $(a, b) \in R \Rightarrow (b, a) \in R$ | every pair has its reverse |
| Transitive | $(a, b), (b, c) \in R \Rightarrow (a, c) \in R$ | every chain is closed |
| Equivalence | all three | splits $A$ into disjoint equivalence classes |

Standard results: "$a - b$ is divisible by $m$" is an equivalence relation on the integers, with $m$ classes. "$a \leq b$" is reflexive and transitive but not symmetric. "Is perpendicular to" on lines is symmetric only. "Is parallel to" is an equivalence relation.

## 4. Counting relations on a set with $n$ elements

| Kind | Number |
|---|---|
| All relations | $2^{n^2}$ |
| Reflexive | $2^{n^2 - n}$ |
| Symmetric | $2^{n(n+1)/2}$ |
| Reflexive and symmetric | $2^{n(n-1)/2}$ |
| Equivalence relations | 1, 2, 5, 15 for $n = 1, 2, 3, 4$ (the number of ways to partition the set) |

**Solved example 2.** $R = \{(1,2),(2,3)\}$ on $A = \{1,2,3\}$. How many ordered pairs must be added to make $R$ an equivalence relation?
Reflexive: add $(1,1),(2,2),(3,3)$. Symmetric: add $(2,1),(3,2)$. Transitive: $(1,2),(2,3)$ needs $(1,3)$, and symmetry then needs $(3,1)$. Total added $= 7$, and $R$ becomes the whole of $A \times A$.

**Method for "smallest equivalence relation containing…" questions.** Join the given elements into groups: every pair given links two elements into the same group. The answer is the sum of (group size)² over all groups, including groups of one element.

**Traps.** A relation can be symmetric and transitive without being reflexive. The empty relation on a non-empty set is symmetric and transitive but not reflexive. When checking transitivity, look for one chain $(a,b),(b,c)$ whose closing pair $(a,c)$ is missing; one failure is enough. Proper subsets exclude the set itself, not the empty set.

**Read in NCERT:** Mathematics Class 11, chapter "Sets", and Class 12 Part 1, chapter "Relations and Functions" (the relations sections only; functions come later in the plan).

# Chemistry · Biomolecules

About one question per paper, taken almost word for word from NCERT. This chapter is memory, not calculation: one careful reading with these notes is enough.

## 1. Carbohydrates

Polyhydroxy aldehydes or ketones, or compounds that give them on hydrolysis.

| Class | On hydrolysis gives | Examples |
|---|---|---|
| Monosaccharides | cannot be hydrolysed further | glucose, fructose, galactose, ribose |
| Oligosaccharides | 2 to 10 monosaccharide units | sucrose, maltose, lactose (all disaccharides) |
| Polysaccharides | many units | starch, cellulose, glycogen |

**Reducing sugars** reduce Tollens' and Fehling's reagents because they have a free aldehyde or ketone group: all monosaccharides, maltose and lactose. **Sucrose is non-reducing.**

## 2. Glucose

An aldohexose, $C_6H_{12}O_6$. Each reaction below proves one feature of its structure:

| Reagent | Product | What it proves |
|---|---|---|
| HI, heat | n-hexane | six carbons in a straight chain |
| HCN or $NH_2OH$ | cyanohydrin or oxime | a carbonyl group |
| Bromine water | gluconic acid | the carbonyl is an aldehyde |
| Acetic anhydride | glucose pentaacetate | five –OH groups |
| Nitric acid | saccharic acid | a primary alcohol group |

**Why a ring structure is needed.** Glucose does not give Schiff's test, does not add $NaHSO_3$, its pentaacetate does not react with $NH_2OH$, and it exists in two forms, α and β. So the –CHO group is not free: it forms a six-membered cyclic hemiacetal (pyranose ring) with the –OH on C5.

- α- and β-glucose differ only in configuration at C1. They are **anomers**, and C1 is the anomeric carbon.
- Fructose is a ketohexose with a five-membered furanose ring; its anomeric carbon is C2.

## 3. Disaccharides and polysaccharides

| Sugar | Made of | Linkage | Reducing? |
|---|---|---|---|
| Sucrose | α-D-glucose + β-D-fructose | C1 of glucose to C2 of fructose | No |
| Maltose | two α-D-glucose | C1 to C4 | Yes |
| Lactose | β-D-galactose + β-D-glucose | C1 of galactose to C4 of glucose | Yes |

- Sucrose is dextrorotatory but its hydrolysis mixture is laevorotatory, so the product is called **invert sugar**.
- **Starch** = amylose (15–20%, unbranched, α-1,4 links, soluble in water) + amylopectin (80–85%, branched, α-1,4 and α-1,6 links, insoluble).

- **Cellulose** = β-D-glucose joined by β-1,4 links. **Glycogen** is like amylopectin but more branched, and is stored in animals.

## 4. Proteins

- Polymers of α-amino acids joined by **peptide bonds**, –CO–NH–. A chain of $n$ amino acids has $n - 1$ peptide bonds.
- Glycine is the only α-amino acid that is not optically active. In water, amino acids exist as zwitterions.
- **Essential** amino acids cannot be made by the body: valine, leucine, isoleucine, lysine, phenylalanine, tryptophan, threonine, methionine, histidine, arginine.

| Structure level | Meaning | Held by |
|---|---|---|
| Primary | sequence of amino acids | peptide bonds |
| Secondary | α-helix or β-pleated sheet | hydrogen bonds |
| Tertiary | overall folding of the chain | hydrogen bonds, disulphide links, van der Waals and electrostatic forces |
| Quaternary | arrangement of two or more chains | the same forces between subunits |

- Fibrous proteins are insoluble (keratin, myosin). Globular proteins are soluble (insulin, albumin).
- **Denaturation** (boiling an egg, curdling of milk) destroys the secondary and tertiary structures. The primary structure stays intact.
- Enzymes are globular proteins that act as specific biological catalysts, for example maltase and invertase.

## 5. Vitamins

Fat-soluble: A, D, E, K. Water-soluble: B group and C (these must be supplied regularly, except $B_{12}$).

| Vitamin | Deficiency disease |
|---|---|
| A | night blindness, xerophthalmia |
| $B_1$ | beriberi |
| $B_2$ | cheilosis |
| $B_6$ | convulsions |
| $B_{12}$ | pernicious anaemia |
| C | scurvy |
| D | rickets, osteomalacia |
| E | muscular weakness, fragile red blood cells |
| K | slow blood clotting |

## 6. Nucleic acids and hormones

- Nucleoside = sugar + base. Nucleotide = sugar + base + phosphate. Nucleotides are joined by phosphodiester links between the 5′ and 3′ carbons.

- **DNA:** 2-deoxyribose; bases A, G, C, T; double helix. A pairs with T by 2 hydrogen bonds, G pairs with C by 3.
- **RNA:** ribose; bases A, G, C, U; single strand; three types: m-RNA, r-RNA, t-RNA.
- **Hormones:** insulin and glucagon are peptides; thyroxine is an iodine-containing amino acid derivative; testosterone, estradiol and progesterone are steroids.

**Traps.** Sucrose is the only common disaccharide that is non-reducing. Lactose contains galactose, not two glucose units. Thymine is in DNA only and uracil in RNA only. Denaturation never breaks the primary structure. Cellulose has β links; starch and glycogen have α links.

**Read in NCERT:** Chemistry Class 12 Part 2, chapter "Biomolecules". Read it line by line; the tables of vitamins and the reactions of glucose are asked directly.